TSP Integer Programming

Review Learning Objectives

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Due Mar 3, 11:59 PM IST

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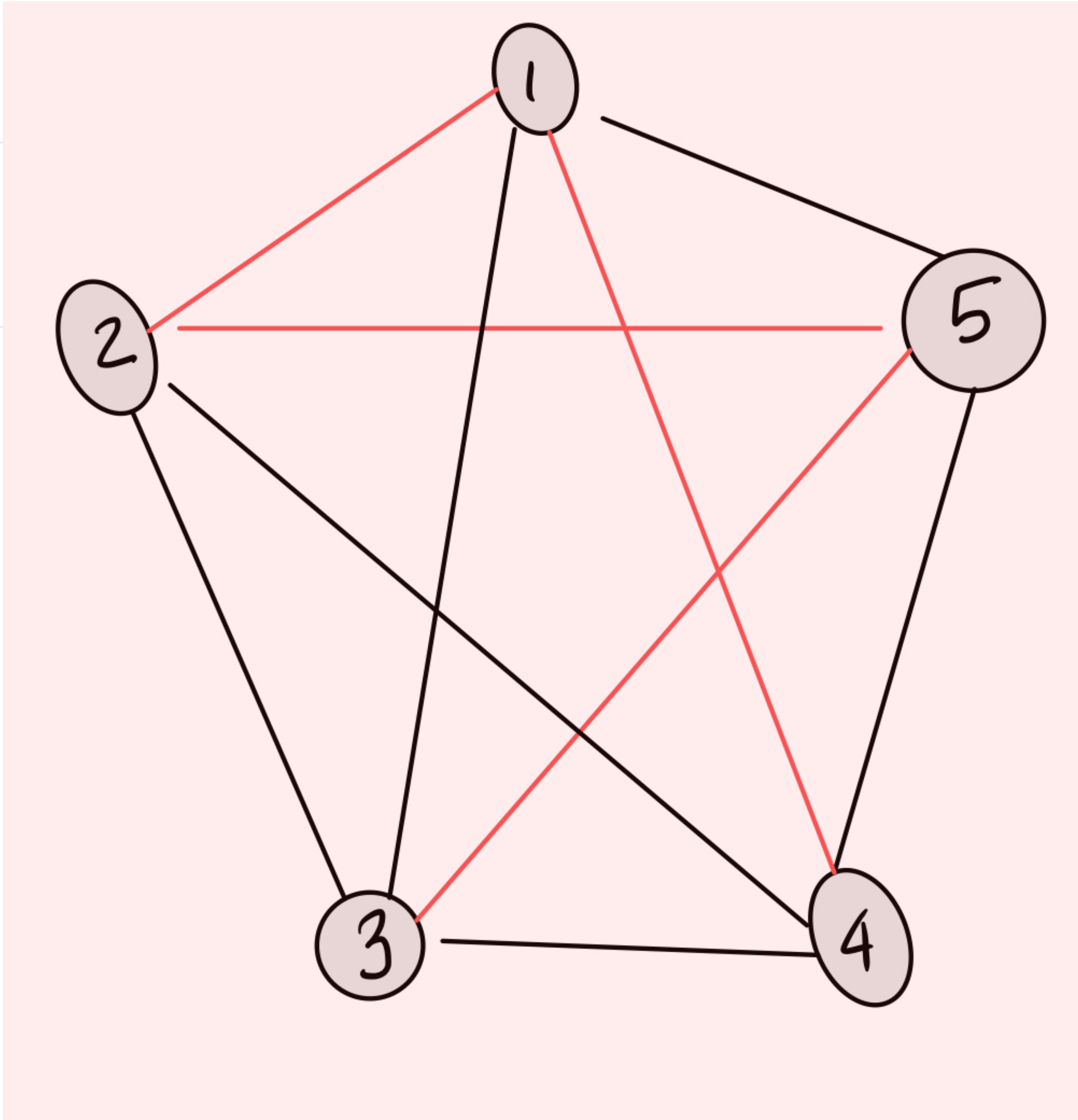
The following problems concern the TSP instance below. Black colored edges have weight 1 whereas red edges have weight 2.



We consider both the encodings MTZ and subtour elimination as presented in our interactive notes. Note that they are more general than the ones in the video which did not go into the details for non-symmetric cases. Please read through these notes and look at the provided code before attempting the problems below.

1. Select all the correct facts about the ILP encoding/decision variables $x_{i,j}$. **2 / 2 points**

- [x] Setting the variable $x_{i,j} = 1$ denotes that we go from vertex i to j using the edge (i, j) in our tour.

  Correct
  Correct

- [ ] The constraint $x_{3,1} + x_{3,2} + x_{3,4} + x_{3,5} \geq 1$ expresses that we have exactly one edge from the tour leaving vertex 3.
- [x] The constraint $x_{2,1} + x_{3,1} + x_{4,1} + x_{5,1} = 1$ expresses that we have exactly one edge entering the vertex 1 in our tour.

  Correct
  Correct

- [ ] Adding the degree constraints and the objective function suffices to find the optimal TSP tour using a binary ILP.

2. Select all the correct facts about the MTZ encoding as it pertains to the problem above. **3 / 4 points**

- [ ] The constraint $t_2 \geq t_3 + x_{2,3}$ is equivalent to the constraint $x_{2,3} = 1 \Rightarrow t_3 \geq t_2 + 1$
- [x] The MTZ encoding allows subtours in the solution but they have to involve the vertex 1.

  This should not be selected
  Incorrect: if we had a subtour involving vertex 1, the vertices not in the subtour will form a subtour not involving vertex 1. The MTZ encoding will rule the latter subtours out and thus all subtours are ruled out. This is explained in detail in our notes.

- [x] Time stamp variables t_2, t_3, t_4, t_5 are added to eliminate possible subtours by assigning increasing time stamps to nodes visited along a tour.

  Correct
  Correct

- [x] The constraint $t_3 \geq t_2 + x_{2,3} - M(1 - x_{2,3})$ for a large number M is equivalent to
  $\textbf{if } (x_{2,3} = 1) \textbf{ then } t_3 \geq t_2 + 1 \textbf{ else } t_3 \geq t_2 - M$

  Correct
  Correct.

3. Suppose we solve the degree ILP relaxation and obtain a subtour $\{2, 3, 5\}$, which of the following inequalities will eliminate the subtour. Once again, please refer to the detailed notes which present the general case for non-symmetric TSPs rather than the video lecture which is restricted to the symmetric case. **2 / 2 points**

- [ ] $x_{2,3} + x_{3,4} + x_{4,5} + x_{5,1} \geq 1$
- [x] $x_{2,4} + x_{3,4} + x_{5,4} + x_{2,1} + x_{3,1} + x_{5,1} \geq 1$

  Correct
  Correct: this states that at least one edge leaving the set $\{2, 3, 4\}$ to a vertex outside of it must be in the tour

- [x] $x_{4,2} + x_{4,3} + x_{4,5} + x_{1,2} + x_{1,3} + x_{1,5} \geq 1$

  Correct
  Correct: this states that at least one edge entering the set $\{2, 3, 4\}$ from outside, must be in the tour